University
Mohammed VI
Polytechnic

College of
Computing

LAB5: Normalization

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering
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Team Information

Team Name	PRINT("MySQL")
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Repository Link	`github.com/abdelillahsaber/DatabaseGroupCC`

1 Task 1

- None. Every relation in the Lab 4 solution is in BCNF.
- For every non-trivial FD, its determinant X is a superkey of the relation.
- For the relations *Advises*, *Attached*, *FundsPrj*, *FundsLab*, *Requires*, and *Reserves*, the Lab 4 constraints imply no non-trivial FDs.
- *Academic*(*pid*) and *NonAcademic*(*pid*) have no non-trivial FDs. These relations therefore satisfy BCNF.

2 Task 2

- **Equipment Inventory:**
  - **Candidate key(s):** $\{serialNo\}$, since $serialNo^+$ = all attributes.
  - **Does the relation satisfy BCNF?** No.
  - **Violating FD:** In 1NF, but not in 3NF or BCNF. The violating FD is $modelId \rightarrow manufacturer$. The closure $\{modelId\}^+ = \{modelId, manufacturer\}$ does not contain all attributes, so *modelId* is not a superkey. The FD is non-trivial and *manufacturer* is non-prime, which also violates 3NF.
  - **Decomposition:** $R_1(modelId, manufacturer)$, $R_2(serialNo, modelId, labId)$
  - **Lossless:** Yes. $R_1 \cap R_2 = \{modelId\}$, which is a primary key of R_1.
  - **FDs preserved:** Yes. $modelId \rightarrow manufacturer$ is in R_1, and $serialNo \rightarrow \{modelId, labId\}$ is in R_2.
- **Project Participation:**
  - **Candidate key(s):** $\{pid, projectCode\}$.Because it determines the whole relation, it is minimal, since $projectCode^+ = \{projectCode, piPid\}$ and $pid^+ = \{pid\}$.
  - **Does the relation satisfy BCNF?** No.
  - **Violating FD:** In 1NF, but not in 3NF or BCNF. The violating FD is $projectCode \rightarrow piPid$. It is non-trivial, *projectCode* is not a superkey, and *piPid* is non-prime. Hence it violates both BCNF and 3NF.
  - **Decomposition:** $R_1(projectCode, piPid)$, $R_2(pid, projectCode, role)$
  - **Lossless:** Yes. $R_1 \cap R_2 = \{projectCode\}$, which is a primary key of R_1.
  - **FDs preserved:** Yes. $projectCode \rightarrow piPid$ is in R_1, and $(pid, projectCode) \rightarrow role$ is in R_2.
- **Equipment Certification Requirement:**
  - **Candidate key(s):** $\{modelId, certCode\}$, because it determines the whole relation, and $certCode^+ = \{certCode, certTitle, issuingAuthority\} \neq R$, and $modelId^+ = \{modelId\}$. So it is minimal.
  - **Does the relation satisfy BCNF?** No.

  - **Violating FD:** In 1NF, but not in 3NF or BCNF. The violating FD is $certCode \rightarrow \{certTitle, issuingAuthority\}$. It is non-trivial and $certCode$ is not a superkey because it does not determine $modelId$. Although $certCode$ is a prime attribute, $certTitle$ and $issuingAuthority$ are non-prime. The dependency therefore violates 3NF as well as BCNF.
  - **Decomposition:** $R_1(certCode, certTitle, issuingAuthority)$, $R_2(modelId, certCode)$
  - **Lossless:** Yes. $R_1 \cap R_2 = \{certCode\}$, which is a key of R_1.
  - **FDs preserved:** Yes. $certCode \rightarrow \{certTitle, issuingAuthority\}$ is in R_1. $R_2(modelId, certCode)$ is all-key: its only candidate key is $\{modelId, certCode\}$, so it has no non-trivial FD to preserve.

- **Supplier Offer:**
  - **Candidate key(s):** $\{suppId, consId, labId\}$. No subset is a superkey.
  - **Does the relation satisfy BCNF?** No.
  - **Violating FD:** In 1NF, but not in 3NF or BCNF. The violating FD is $suppId \rightarrow supplierName$. The closure $\{suppId\}^+ = \{suppId, supplierName\}$ is not the full schema, so $suppId$ is not a superkey. The FD is non-trivial and $supplierName$ is non-prime, which also violates 3NF.
  - **Decomposition:** $R_1(suppId, supplierName)$, $R_2(suppId, consId, labId, unitPrice)$
  - **Lossless:** Yes. $R_1 \cap R_2 = \{suppId\}$, which is a primary key of R_1.
  - **FDs preserved:** Yes. $suppId \rightarrow supplierName$ is in R_1, and $(suppId, consId, labId) \rightarrow unitPrice$ is in R_2.

- **Reservation and Project Information:**
  - **Candidate key(s):** $\{resId\}$. $resId^+ = \{resId, projectCode, madeByPid, status\}$, plus $projectTitle$ via $projectCode \rightarrow projectTitle$, which gives all attributes.
  - **Does the relation satisfy BCNF?** No.
  - **Violating FD:** In 1NF, but not in 3NF or BCNF. The violating FD is $projectCode \rightarrow projectTitle$. The closure $\{projectCode\}^+ = \{projectCode, projectTitle\}$ does not contain all attributes, so $projectCode$ is not a superkey. The FD is non-trivial and $projectTitle$ is non-prime, which also violates 3NF.
  - **Decomposition:** $R_1(projectCode, projectTitle)$, $R_2(resId, projectCode, madeByPid, status)$
  - **Lossless:** Yes. $R_1 \cap R_2 = \{projectCode\}$, which is a primary key of R_1.
  - **FDs preserved:** Yes. $projectCode \rightarrow projectTitle$ is in R_1, and $resId \rightarrow \{projectCode, madeByPid, status\}$ is in R_2.

3 The role of each team member

All team members contributed equally to the work. We discussed all questions together for each relation, including its functional dependencies, candidate keys, normal forms, and decomposition. We also decided how to structure and present the report.

4 The key challenges encountered and how they were resolve

Our main question was whether we needed to discuss 2NF when identifying the strongest normal form satisfied by each relation. The professor's email to the class clarified this requirement and resolved our confusion.

5 Any assumptions made

We assumed that all attribute values are atomic, so the original relations satisfy 1NF. We did not assume additional uniqueness constraints on names or titles.